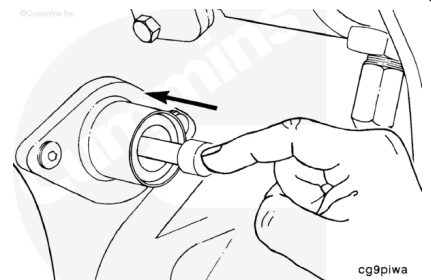


Trainings ▾

Preparatory Steps

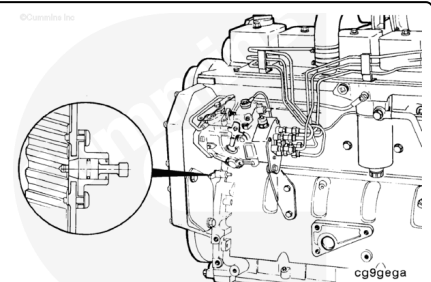
Locate top dead center for cylinder number 1 by barring crankshaft slowly while pressing on engine timing pin.

Service Tip: If the timing pin has been damaged and can **not** be used to locate top dead center, use the following steps to locate top dead center (TDC)..



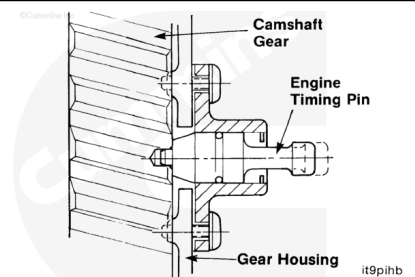
Remove

Remove the timing pin assembly and seal.



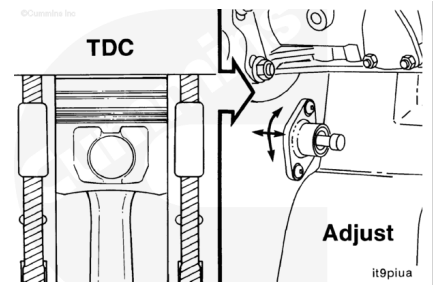
Install

The timing pin assembly is precisely located on the gear housing to correspond to Top Dead Center (TDC) for cylinder number 1.

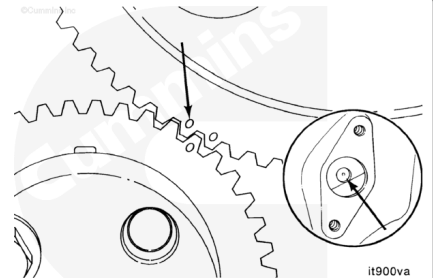


If a gear housing other than the original housing is installed, the timing pin assembly must be accurately aligned to correspond to TDC for cylinder number 1.

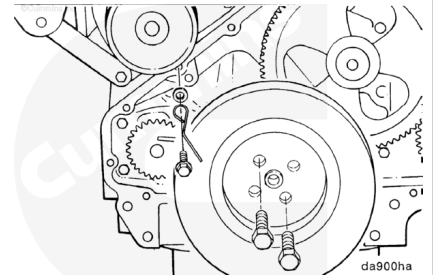
Failure to locate the timing pin assembly to TDC for cylinder number 1 will result in incorrect injection timing.



Verify that the number 1 cylinder is at or near TDC on the compression stroke by rotating the crankshaft until the engine timing pin engages in the camshaft gear hole or the hole is visible through the gear housing. Disengage the timing pin.

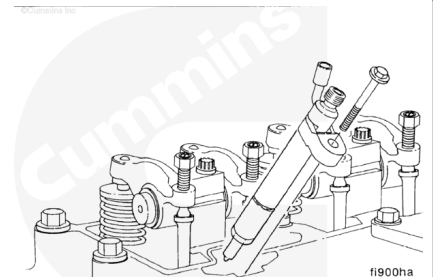


Fabricate and install a wire pointer for the front of the engine. This can be done by forming a piece of wire that can be tightened under one of the gear housing capscrews. The wire **must** extend from the gear cover to a place on the crankshaft vibration damper that is easily seen.



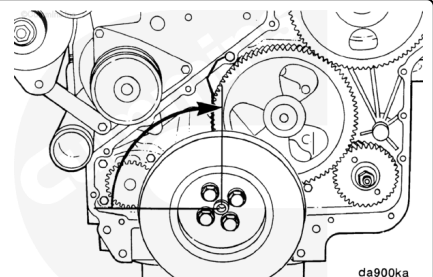
Remove all injectors. Refer to Procedure 006-026 in Section 6. ([/qs3/pubsys2/xml/en/procedures/41/41-006-026.html](http://qs3/pubsys2/xml/en/procedures/41/41-006-026.html))

This step is important to release cylinder pressure so the crankshaft can be rotated smoothly to locate TDC for cylinder number 1.



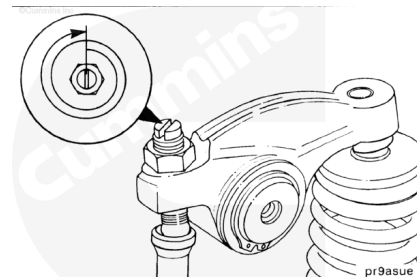
Use tool, Part Number 3377371, Engine Barring Gear.

Rotate the crankshaft one-quarter revolution in the direction of normal engine rotation.



Tighten the adjusting capscrew for the number 1 intake valve to zero (0) lash plus 5 turns.

Note : Leave the adjusting screw in this position until TDC is established.

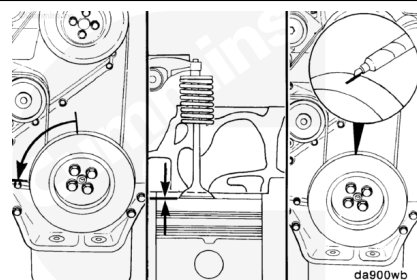


⚠ CAUTION ⚠

Use extreme care when rotating the crankshaft. Use of too much force could damage the valve or push rod.

Rotate the crankshaft slowly in the opposite direction of normal engine rotation until the piston touches the intake valve.

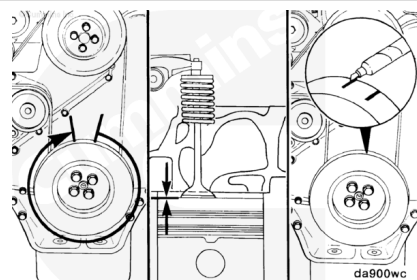
Mark the vibration damper at the wire pointer.



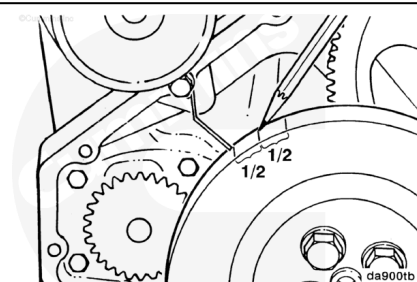
⚠ CAUTION ⚠

Make sure that the piston touches the intake valve with approximately the same amount of force as in the previous step.

- Rotate the crankshaft in the direction of normal engine rotation until the piston touches the intake valve.
- Mark the vibration damper at the wire pointer.

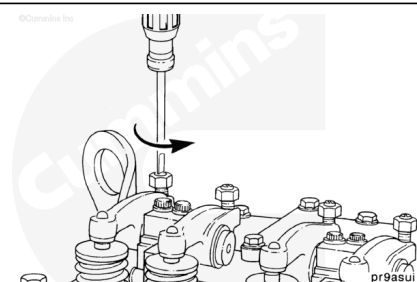


Measure the distance and mark the vibration damper at one-half the distance between the two marks. This mark is the TDC mark.

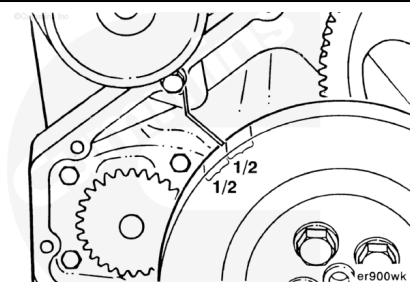


⚠ CAUTION ⚠

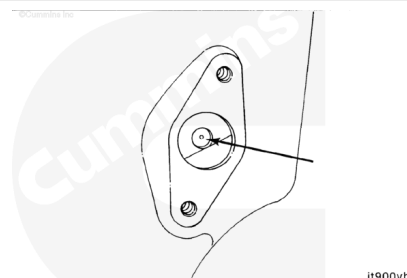
Completely loosen the intake valve adjusting screw. Failure to do so will result in damage to the intake valve or push rod when the crankshaft is rotated.



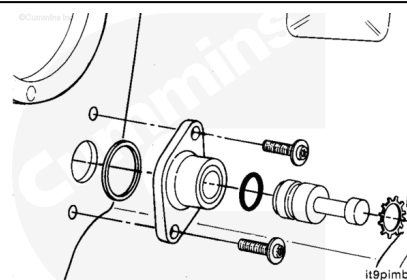
Rotate the crankshaft in the direction of normal engine rotation until the pointer is aligned with the TDC mark; then rotate the crankshaft one additional revolution.



The timing pin hole in the camshaft gear **must** be visible, or felt through the back side of the gear housing. If not, the crankshaft **must** be rotated one revolution in the direction of engine rotation.



Install the o-ring into the groove on the timing pin assembly.
Lubricate the o-ring with 15W-40 engine oil.



Push the pin into the hole in the camshaft gear to align the housing.

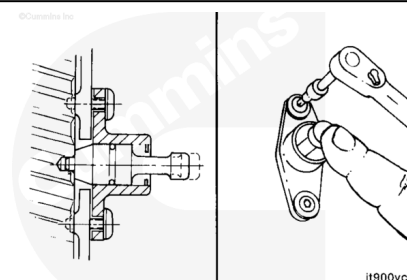
Hold the pin while tightening the Torx™ capscrews.

Torque Value:

Timing Pin Torx™ Capscrew

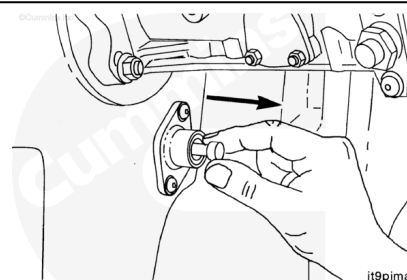
1.8 n•m [71 in-lb]

Check the pin to be sure it did **not** slip out of the camshaft gear hole when the capscrews were tightened.



⚠ CAUTION ⚠

To prevent damage to the pin and gear, be sure the timing pin is disengaged before rotating the crankshaft.



Finishing Steps

- Install the gear cover. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/100/100-001-031-tr.html>)
- Install the vibration damper.
 - Use this procedure for rubber vibration damper. Refer to Procedure 001-051 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-051-tr.html>)
 - Use this procedure for viscous vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-052-tr.html>)
- Remove the wire pointer.
- Adjust overhead. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/41/41-003-004.html>)
- Install all injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/41/41-006-026.html>)
- Operate the engine and check for leaks.

